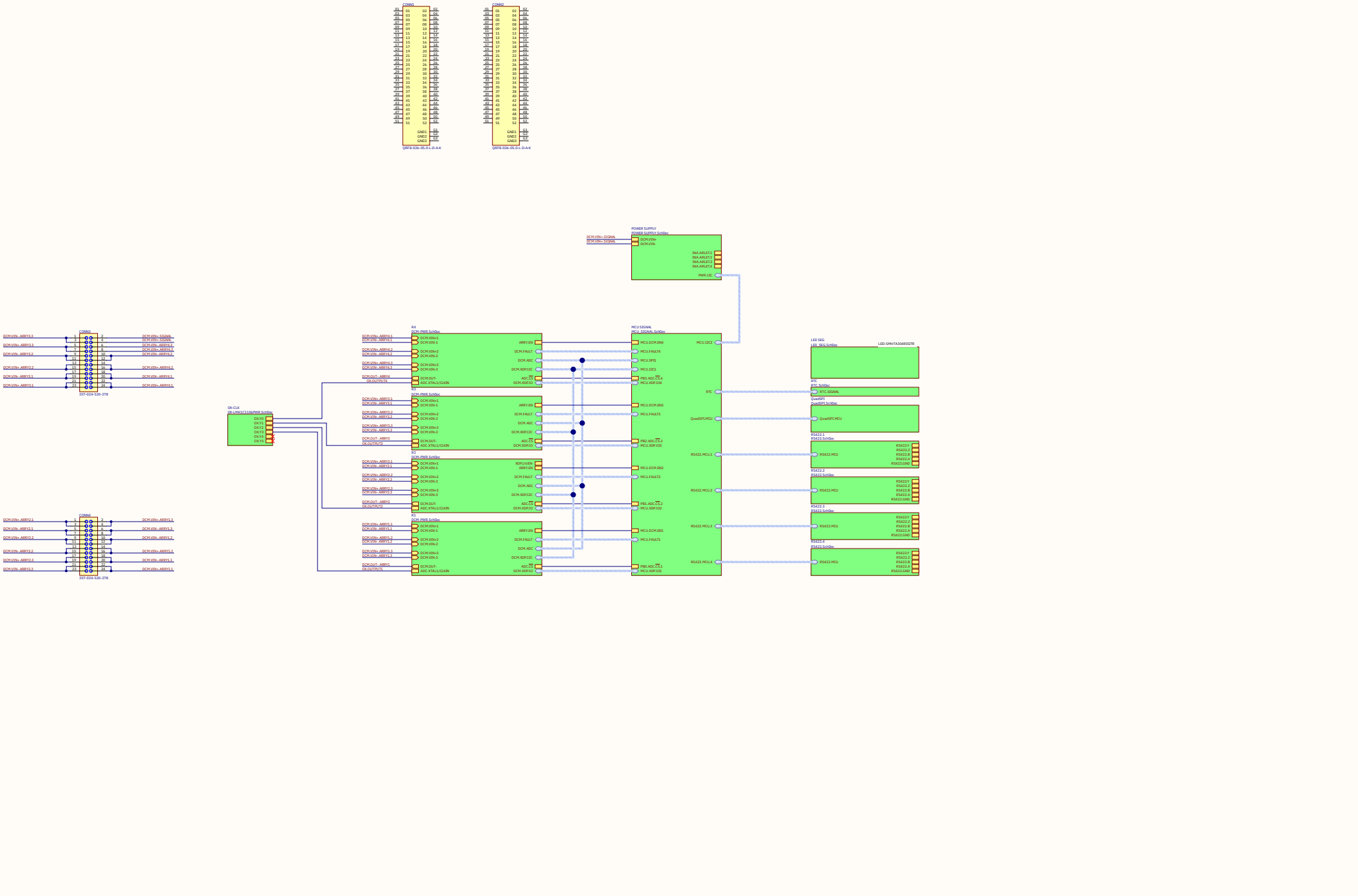
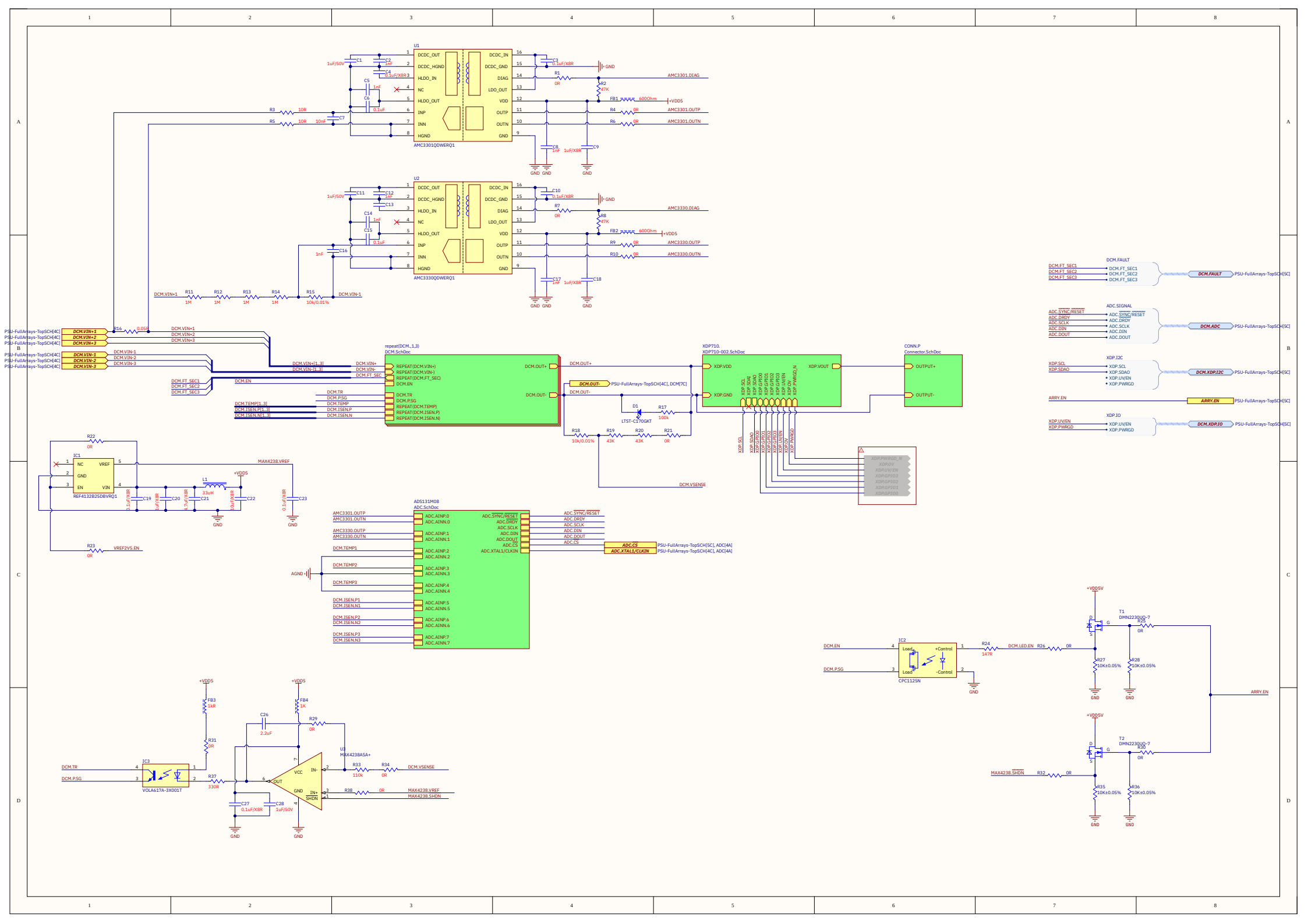
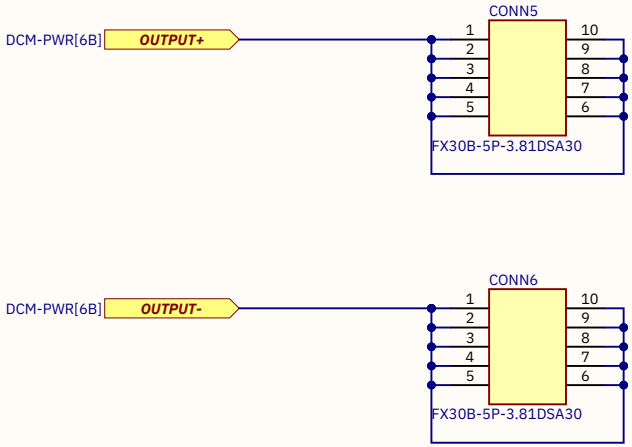
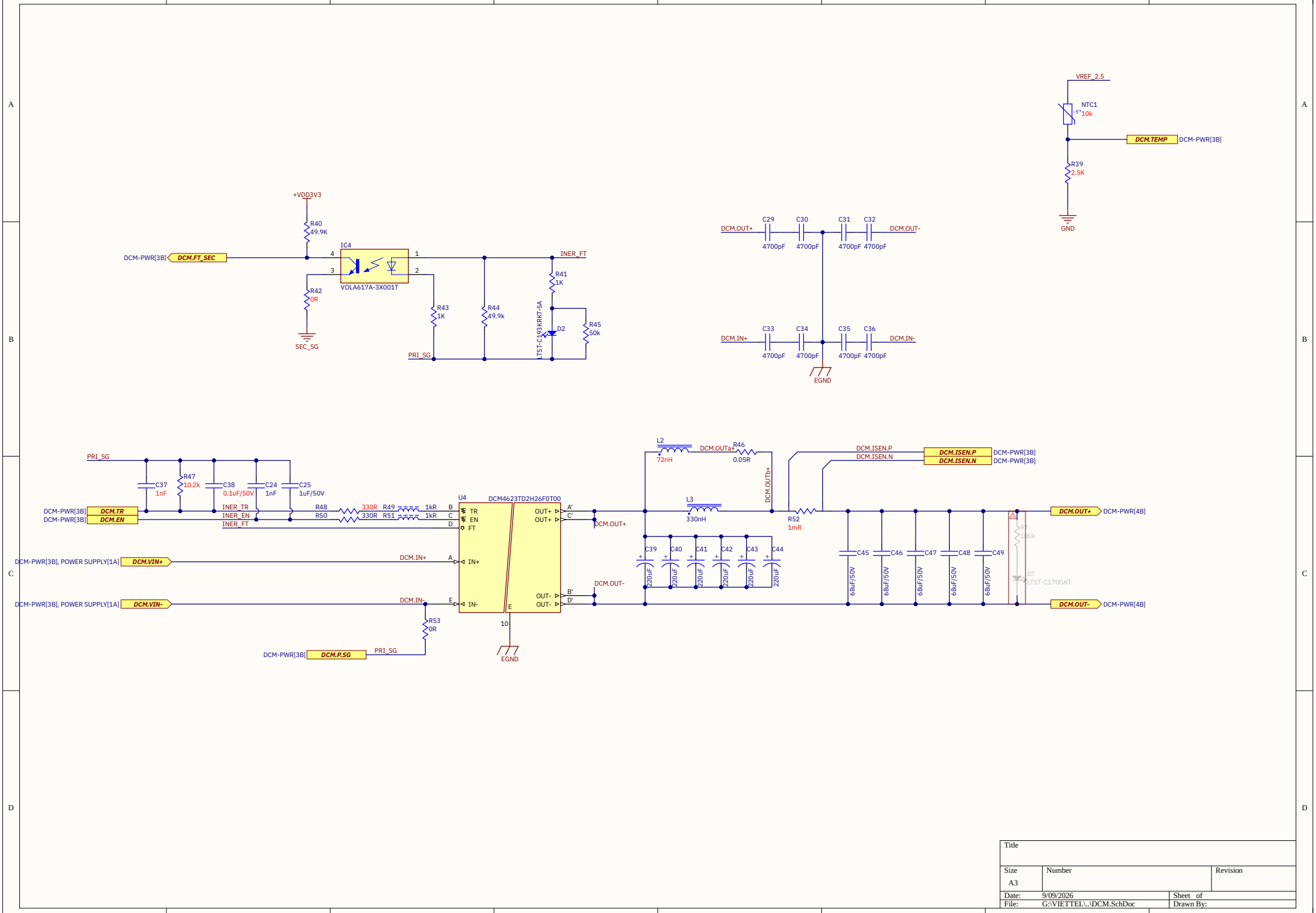


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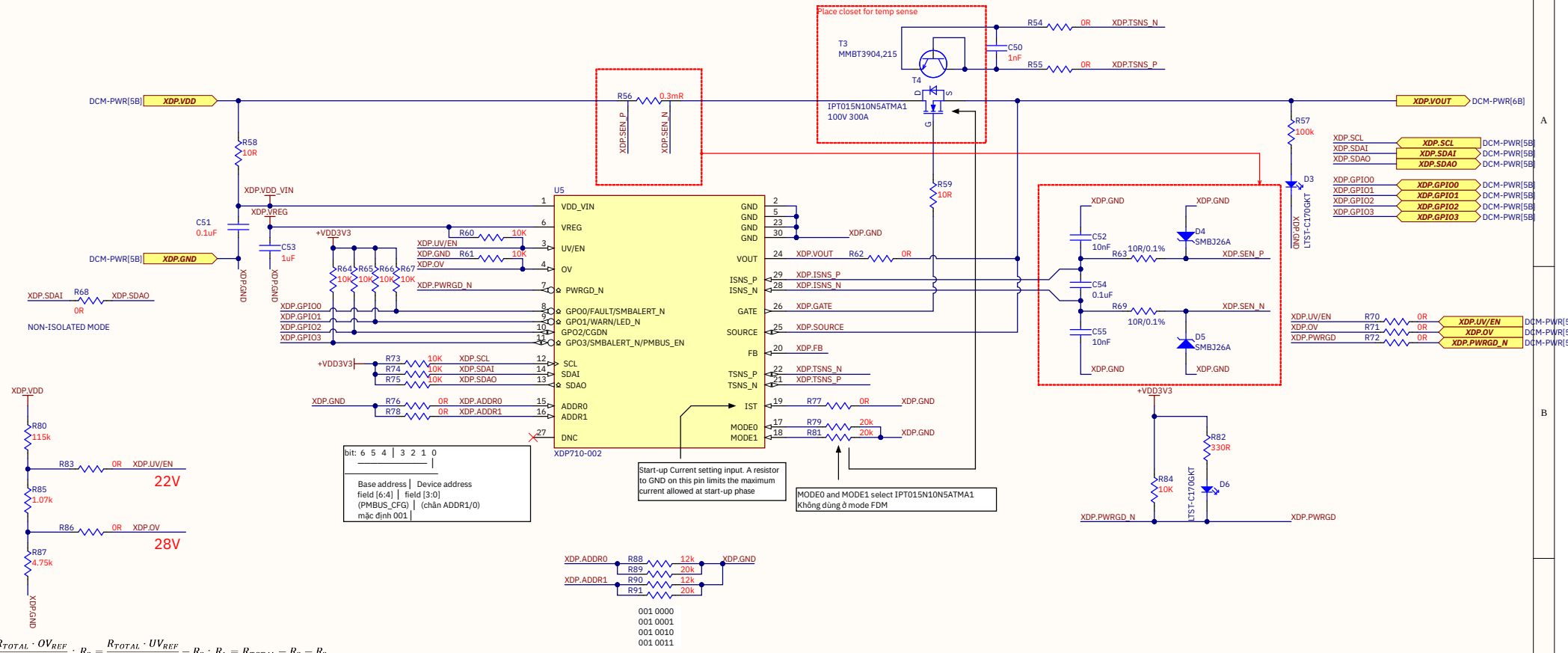

U1
AMC3301QDWERQ1
DCDC_OUT
DCDC_HGND
HLDO_IN
NC
HLDO_OUT
INP
INN
HGND
DCDC_IN
DCDC_GND
DIAG
LDO_OUT
VDD
OUTP
OUTN
GND
AMC3301.DIAG
AMC3301.OUTP
AMC3301.OUTN
U2
AMC3330QDWERQ1
AMC3330.DIAG
AMC3330.OUTP
AMC3330.OUTN
DCM.VIN+1
DCM.VIN-1
repeat(DCM.,1,3)
DCM.SchDoc
XDP710.
XDP710-002.SchDoc
CONN.P
Connector.SchDoc
DCM.OUT+
DCM.OUT-
DCM.VSENSE
DCM.EN
DCM.TR
DCM.PSG
DCM.TEMP
DCM.ISEN.P
DCM.ISEN.N
DCM.FAULT
DCM.ADC
DCM.XDP.I2C
DCM.XDP.IO
ARRY.EN
ADC.SIGNAL
ADS131M08
ADC.SchDoc
ADC.CS
ADC.XTAL1/CLKIN
MAX4238.VREF
MAX4238.SHDN
VREF2V5.EN
REF4132B2SDBVRQ1
VOLA617A-3X001T
CPC1125N
DMN2230UQ-7
+VDD5
AGND
GND



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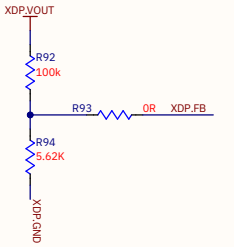


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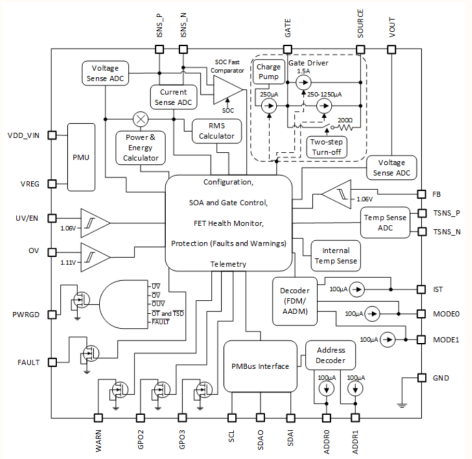
$$R_3 = \frac{R_{TOTAL} \cdot OV_{REF}}{V_OV} \; ; \; R_2 = \frac{R_{TOTAL} \cdot UV_{REF}}{V_UV} - R_3 \; ; \; R_1 = R_{TOTAL} - R_2 - R_3$$

với OV_REF = V_OV_UTH = 1,11 V và UV_REF = V_UVEN_LTH = 1,06 V (Table 9).



$$R_5 = \frac{V_{REF} \cdot R_4}{OUV - V_{REF}} = \frac{1{,}11 \times 100000}{21 - 1{,}11} = 5581\ \Omega$$

Thông số	Giá trị
Ngưỡng nhả / phục hồi (V_FB_UTH = 1,11 V)	**20,86 V**
Ngưỡng tác động lỗi OUV (V_FB_LTH = 1,06 V)	**19,92 V**
Worst-case (R ±1% + chuẩn IC) — nhả	20,10 – 21,64 V
Worst-case — tác động	19,18 – 20,69 V
Dòng lấy từ VOUT @24 V	227 µA (5,5 mW)
Áp chân FB @24 V	1,28 V



3.6 Electrical characteristics

Table 9 Electrical characteristics

VDD_VIN - GND = 48 V, V_{ISNS_P} = VDD_VIN, $V_{\Delta ISNS}$ = (V_{ISNS_P} - V_{ISNS_N}) = 0 V, T_J = –40°C to +125°C, unless otherwise noted.

Parameter	Symbol	Values			Unit	Note or condition
		Min.	Typ.	Max.		
UV/EN, OV and FB in AADM						
Input upper threshold	V_{UVEN_UTH}, V_{OV_UTH}, V_{FB_UTH}	1.09	1.11	1.13	V	

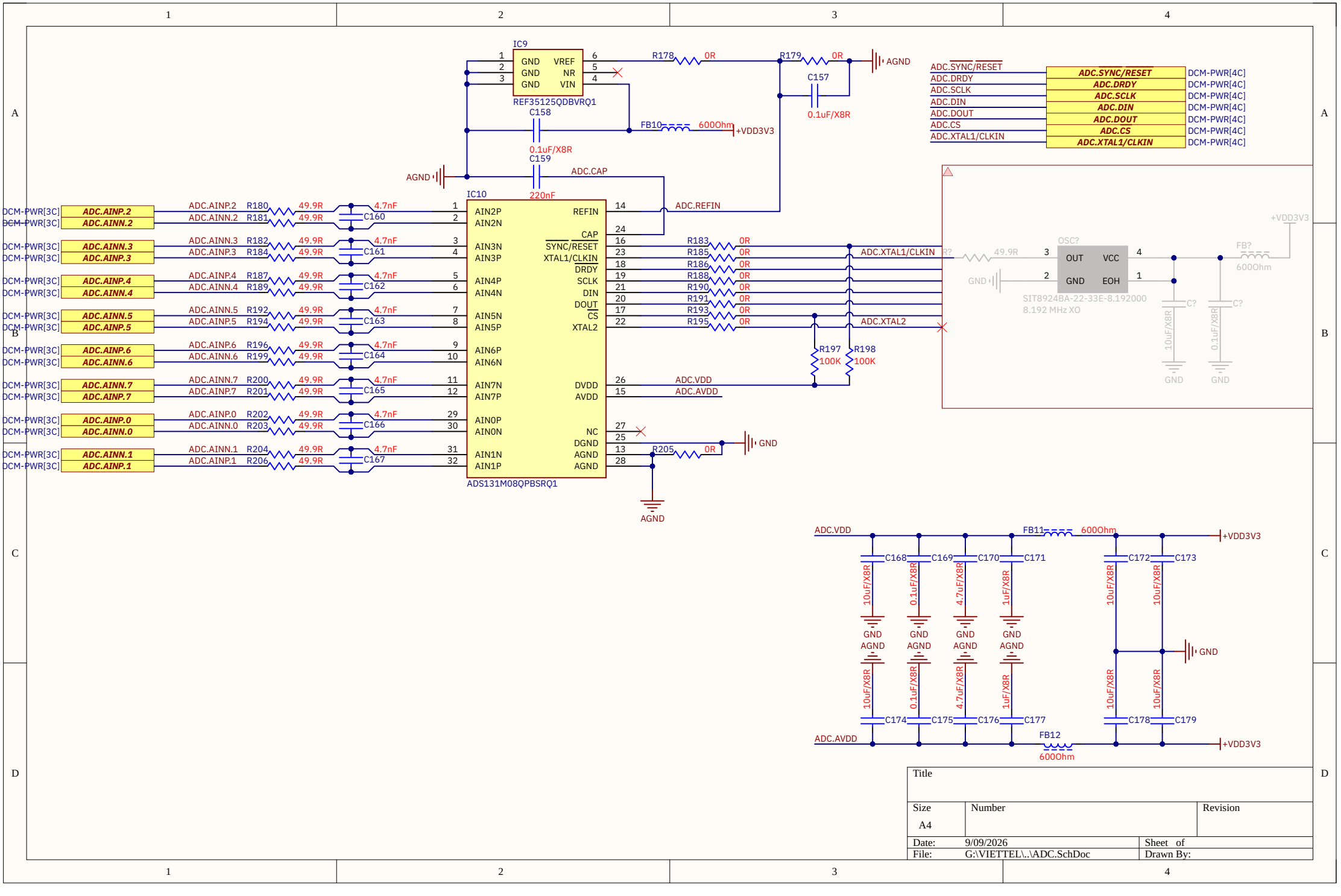
(table continues...)

Table 9 (continued) Electrical characteristics

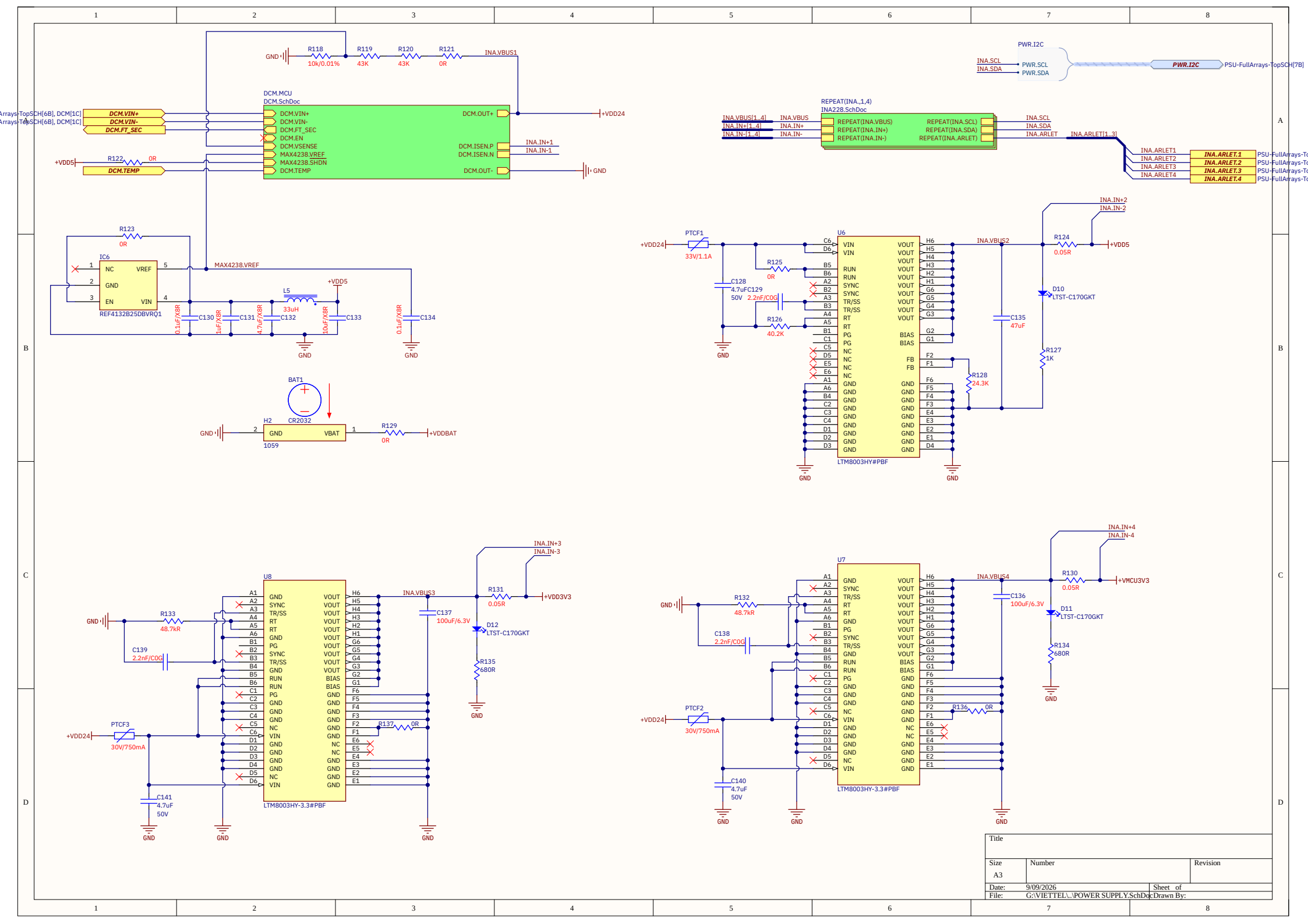
VDD_VIN - GND = 48 V, V_{ISNS_P} = VDD_VIN, $V_{\Delta ISNS}$ = (V_{ISNS_P} - V_{ISNS_N}) = 0 V, T_J = –40°C to +125°C, unless otherwise noted.

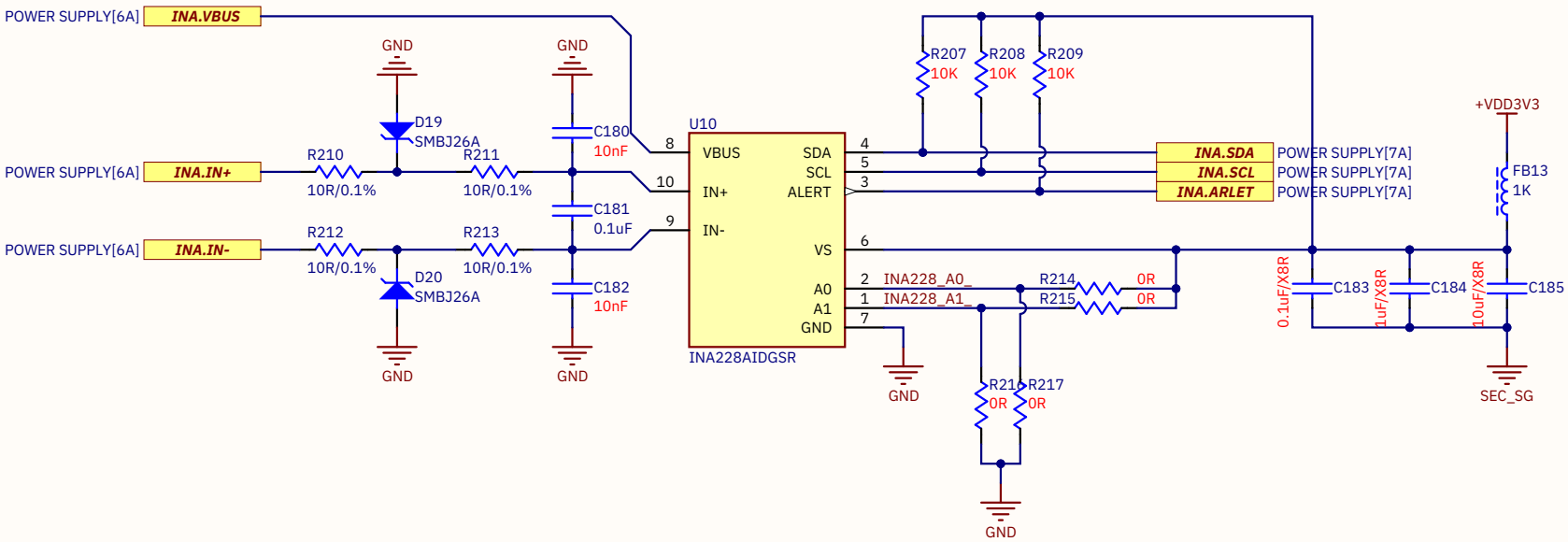
Parameter	Symbol	Values			Unit	Note or condition
		Min.	Typ.	Max.		
Input lower threshold	V_{UVEN_LTH}, V_{OV_LTH}, V_{FB_LTH}	1.04	1.06	1.08	V	

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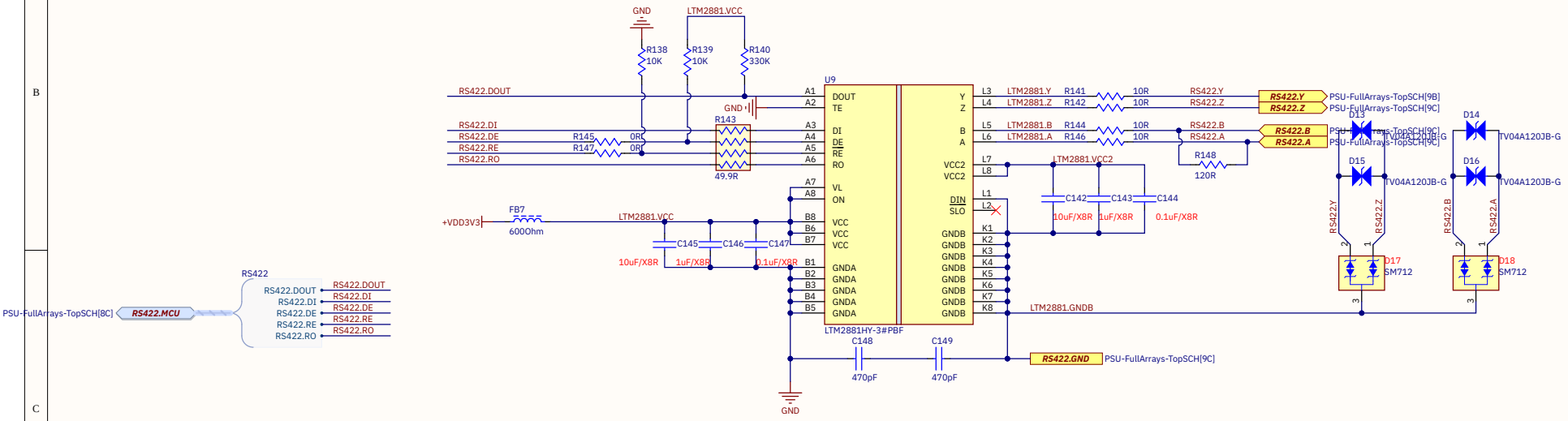


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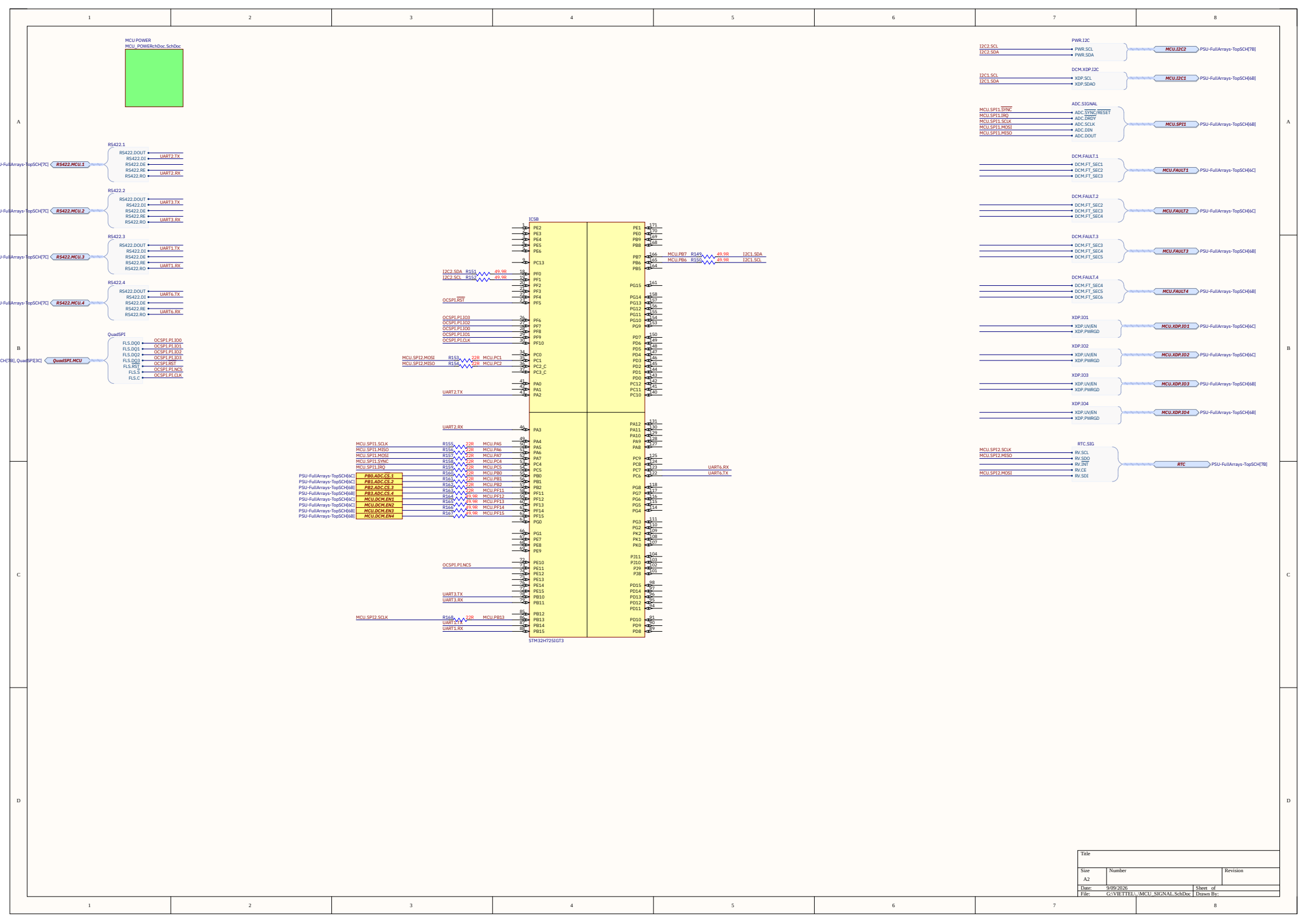

DCM.MCU
DCM.SchDoc
DCM.VIN+
DCM.VIN-
DCM.FT_SEC
DCM.TEMP
DCM.EN
DCM.VSENSE
MAX4238.VREF
MAX4238.SHDN
DCM.OUT+
DCM.OUT-
DCM.ISEN.P
DCM.ISEN.N
R118 10K/0.01%
R119 43K
R120 43K
R121 0R
R122 0R
INA.VBUS1
INA.IN+1
INA.IN-1
+VDD24
+VDD5
GND
PWR.I2C
INA.SCL
INA.SDA
PWR.SCL
PWR.SDA
REPEAT(INA.,1,4)
INA228.SchDoc
REPEAT(INA.VBUS)
REPEAT(INA.IN+)
REPEAT(INA.IN-)
REPEAT(INA.SCL)
REPEAT(INA.SDA)
REPEAT(INA.ARLET)
INA.ARLET[1..3]
INA.ARLET1
INA.ARLET2
INA.ARLET3
INA.ARLET4
R123 0R
IC6
REF4132B25DBVRQ1
MAX4238.VREF
L5 33uH
C130 0.1uF/X8R
C131 1uF/X8R
C132 4.7uF/X8R
C133 10uF/X8R
C134 0.1uF/X8R
BAT1
H2
CR2032
1059
R129 0R
+VDDBAT
PTCF1 33V/1.1A
U6
LTM8003HY#PBF
C128 4.7uF 50V
C129 2.2nF/C0G
R125 0R
R126 40.2K
INA.VBUS2
R124 0.05R
+VDD5
INA.IN+2
INA.IN-2
D10 LTST-C170GKT
C135 47uF
R127 1K
R128 24.3K
U8
LTM8003HY-3.3#PBF
R133 48.7kR
C139 2.2nF/C0G
PTCF3 30V/750mA
C141 4.7uF 50V
INA.VBUS3
C137 100uF/6.3V
R131 0.05R
+VDD3V3
INA.IN+3
INA.IN-3
D12 LTST-C170GKT
R135 680R
R137 0R
U7
LTM8003HY-3.3#PBF
R132 48.7kR
C138 2.2nF/C0G
PTCF2 30V/750mA
C140 4.7uF 50V
INA.VBUS4
C136 100uF/6.3V
R130 0.05R
+VMCU3V3
INA.IN+4
INA.IN-4
D11 LTST-C170GKT
R134 680R
R136 0R
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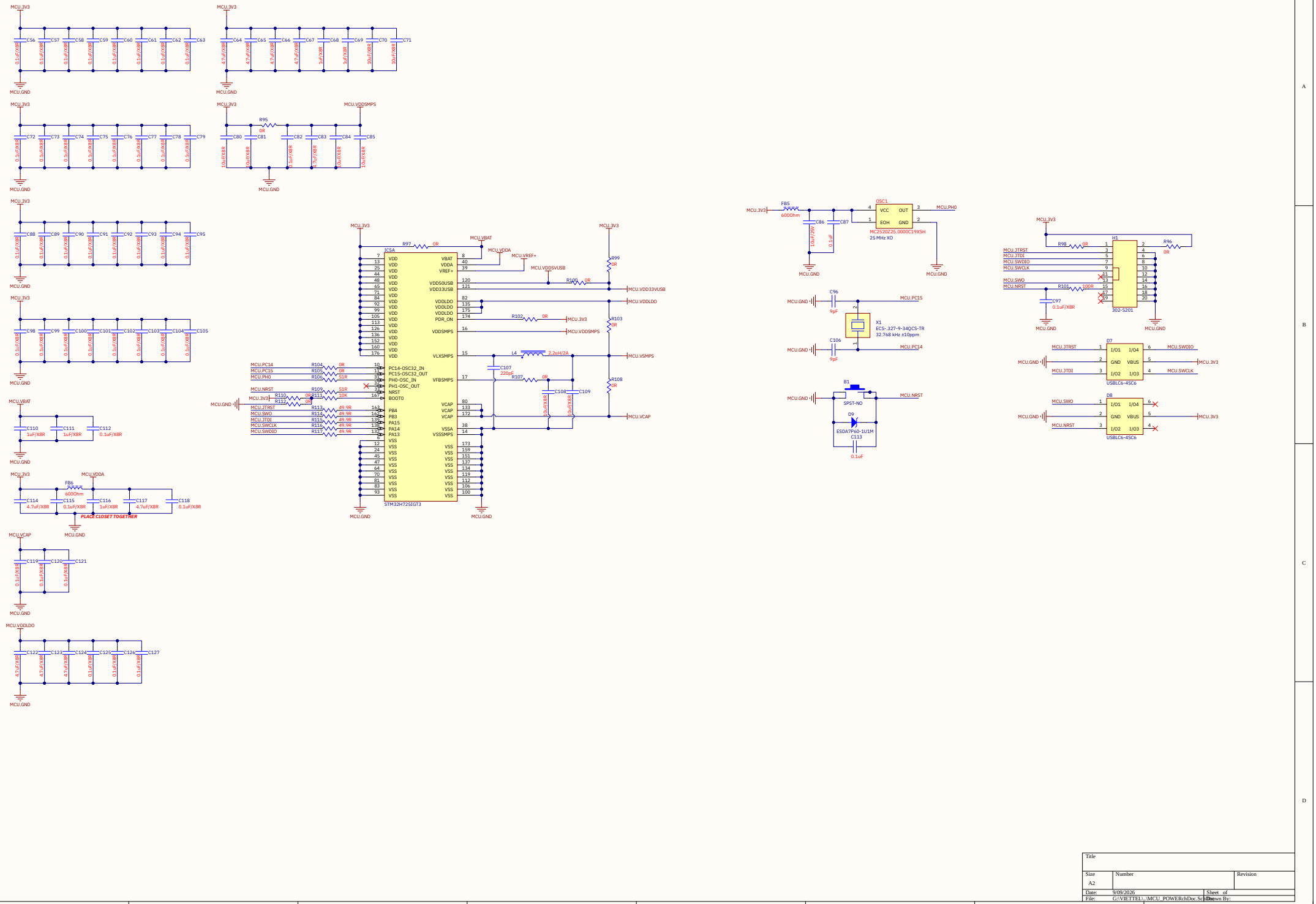


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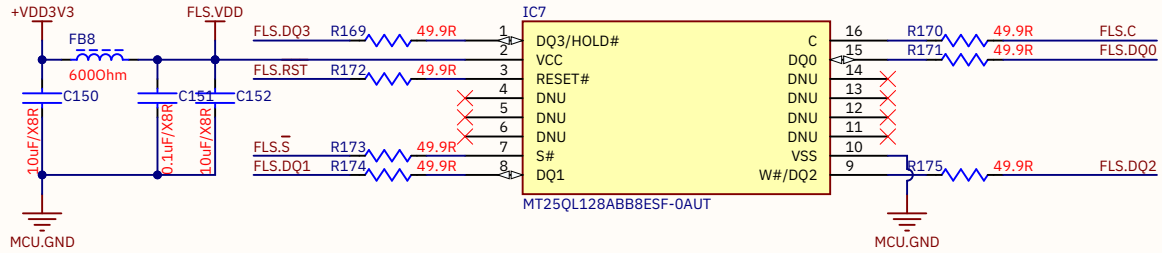


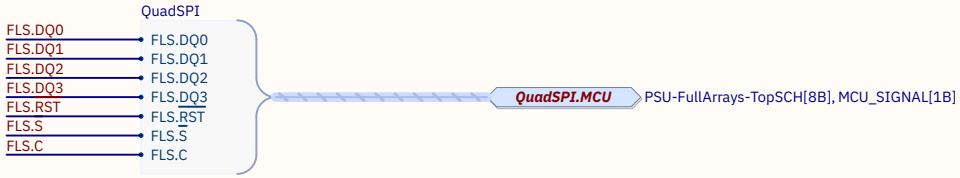
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MCU.POWER
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RS422.1
RS422.DOUT
RS422.DI
RS422.DE
RS422.RE
RS422.RO
UART2.TX
UART2.RX
RS422.MCU.1
RS422.2
UART3.TX
UART3.RX
RS422.MCU.2
RS422.3
UART1.TX
UART1.RX
RS422.MCU.3
RS422.4
UART6.TX
UART6.RX
RS422.MCU.4
QuadSPI
FLS.DQ0
FLS.DQ1
FLS.DQ2
FLS.DQ3
FLS.RST
FLS.S
FLS.C
OCSPI.P1.IO0
OCSPI.P1.IO1
OCSPI.P1.IO2
OCSPI.P1.IO3
OCSPI.RST
OCSPI.P1.NCS
OCSPI.P1.CLK
QuadSPI.MCU
U5B
STM32H725IGT3
I2C2.SCL
I2C2.SDA
I2C1.SDA
I2C1.SCL
OCSPI.RST
OCSPI.P1.IO3
OCSPI.P1.IO2
OCSPI.P1.IO0
OCSPI.P1.IO1
OCSPI.P1.CLK
MCU.SPI2.MOSI
MCU.SPI2.MISO
UART2.TX
UART2.RX
MCU.SPI1.SCLK
MCU.SPI1.MISO
MCU.SPI1.MOSI
MCU.SPI1.SYNC
MCU.SPI1.IRQ
PB0.ADC.CS.1
PB1.ADC.CS.2
PB2.ADC.CS.3
PB3.ADC.CS.4
MCU.DCM.EN1
MCU.DCM.EN2
MCU.DCM.EN3
MCU.DCM.EN4
OCSPI.P1.NCS
UART3.TX
UART3.RX
MCU.SPI2.SCLK
UART1.TX
UART1.RX
UART6.RX
UART6.TX
PWR.I2C
PWR.SCL
PWR.SDA
MCU.I2C2
DCM.XDP.I2C
XDP.SCL
XDP.SDAO
MCU.I2C1
ADC.SIGNAL
ADC.SYNC/RESET
ADC.DRDY
ADC.SCLK
ADC.DIN
ADC.DOUT
MCU.SPI1
DCM.FAULT.1
DCM.FT_SEC1
DCM.FT_SEC2
DCM.FT_SEC3
MCU.FAULT1
DCM.FAULT.2
DCM.FT_SEC4
MCU.FAULT2
DCM.FAULT.3
DCM.FAULT.4
DCM.FT_SEC5
DCM.FT_SEC6
MCU.FAULT3
MCU.FAULT4
XDP.IO1
XDP.UV/EN
XDP.PWRGD
MCU.XDP.IO1
XDP.IO2
MCU.XDP.IO2
XDP.IO3
MCU.XDP.IO3
XDP.IO4
MCU.XDP.IO4
RTC.SIG
RV.SCL
RV.SDO
RV.INT
RV.CE
RV.SDI
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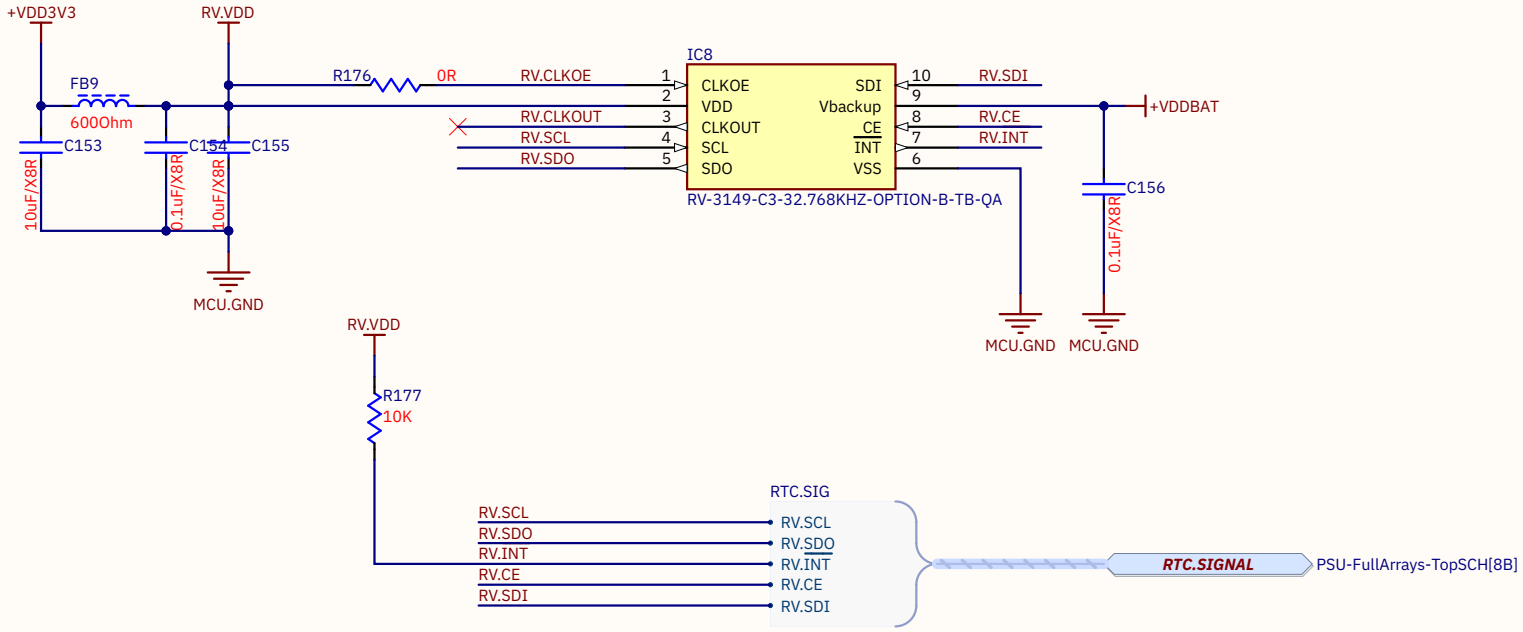


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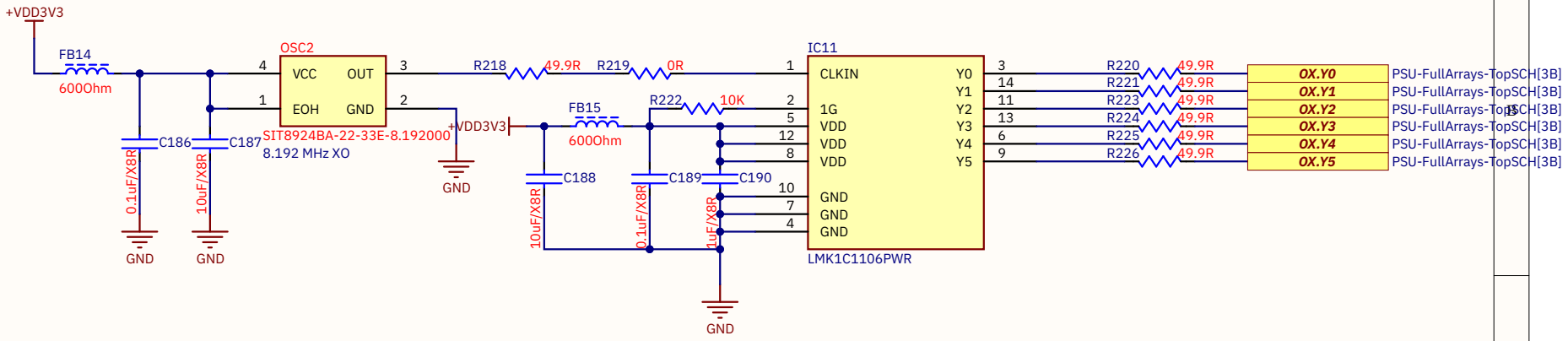


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Board Stack Report